

CBSE Class 12 Pe — Test and Measurement in Sports

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark (MCQ/Assertion-Reason/VSA)

- Q1.** Who developed the Harvard Step Test? (A) Brouha (B) Miller (C) Coubertin (D) James [PYQ 2024] | Confidence: Very High
- Q2.** In the Rikli & Jones Senior Citizen Fitness Test, what is the weight of the dumbbell used for women in the Arm Curl Test? (A) 4 pounds (B) 5 pounds (C) 8 pounds (D) 10 pounds [PYQ 2024] | Confidence: Very High
- Q3.** Which of the following is the correct formula to calculate the Fitness Index in the Harvard Step Test (Long Method)? (A) $\frac{100 \times \text{Test duration in seconds}}{2 \times \text{Sum of heart beats in recovery periods}}$ (B) $\frac{100 \times \text{Test duration in minutes}}{2 \times \text{Sum of heart beats in recovery periods}}$ (C) $\frac{100 \times \text{Test duration in seconds}}{5.5 \times \text{Pulse count of 1-1.5 Min}}$ (D) $\frac{2 \times \text{Test duration in seconds}}{100 \times \text{Sum of heart beats}}$ [PYQ 2024] | Confidence: Very High
- Q4.** In the SAI Khelo India Fitness Test battery for the 5-8 years age group, which test is used to measure static balance? (A) Plate Tapping Test (B) 50m Dash (C) Flamingo Balance Test (D) Partial Curl-up [PYQ 2024] | Confidence: Very High
- Q5.** The '8-foot up & go test' in the Senior Citizen Fitness Test battery is designed to measure: (A) Agility and dynamic balance (B) Upper body flexibility (C) Cardiovascular endurance (D) Lower body strength [PYQ 2024] | Confidence: Very High
- Q6.** In the Partial Curl-up test, what is the standard distance between the two parallel lines on the mat? (A) 8 inches (B) 4 inches (C) 6 inches (D) 10 inches [PYQ 2024] | Confidence: Very High
- Q7.** The 'Chair Sit and Reach Test' for senior citizens is specifically done to check: (A) Lower body flexibility (B) Upper body flexibility (C) Cardiovascular endurance (D) Agility [PYQ 2024] | Confidence: Very High
- Q8.** Which of the following tests is used to measure cardiovascular endurance in the SAI Khelo India battery (9-18 years)? (A) 600m Run/Walk (B) 50m Dash (C) Sit and Reach Test (D) Partial Curl-up [PYQ] | Confidence: Very High
- Q9.** What is the primary purpose of the 4 × 10m Shuttle Run? (A) To measure speed and agility (B) To measure lower body strength (C) To measure cardiovascular endurance (D) To measure flexibility [PYQ] | Confidence: High
- Q10.** What is the dimension of the layout mat in the Johnsen-Methney Test of Motor Educability? (A) 15 × 3 feet (B) 15 × 3 meters (C) 15 × 2 feet (D) 15 × 2 meters [PYQ 2025] | Confidence: High
- Q11.** What is the standard height of the bench used in the Harvard Step Test for women? (A) 45 cm (16 inches) (B) 50 cm (20 inches) (C) 40 cm (D) 55 cm [PYQ] | Confidence: Very High
- Q12.** In which test is the score recorded to the nearest centimetre between the initial and final score to measure flexibility? (A) Partial curl-up (B) Back scratch test (C) Sit and reach test (D) Chair sit and reach test [PYQ 2025] | Confidence: High
- Q13.** How many times are pulse rates taken to estimate the fitness level in the Harvard Step Test (Long formula)? (A) 3 times (B) 2 times (C) 5 times (D) 4 times [PYQ] | Confidence: Very High
- Q14.** Match the following test items with what they measure: I. Plate Tapping Test 1. Upper body strength boys II. Push up 2. Reaction time / coordination III. Partial Curl up 3. Upper body strength girls IV. Modified pushup 4. Abdominal strength (A) I-2, II-1, III-4, IV-3 (B) I-2, II-3, III-1, IV-4 (C) I-1, II-3, III-2, IV-4 (D) I-2, II-3, III-4, IV-1 [PYQ 2024] | Confidence: Very High
- Q15.** Given below are two statements labeled Assertion (A) and Reason (R): Assertion (A): Body Mass Index (BMI) is widely used by physical education teachers to measure an athlete's cardiovascular endurance. Reason (R): BMI is a statistical value calculated using an individual's height and weight to categorize their body weight. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. [PYQ 2024] | Confidence: High
- Q16.** The 'Chair Stand Test' for senior citizens is designed to evaluate: (A) Lower body strength (B) Upper body strength (C) Aerobic fitness (D) Anaerobic fitness [PYQ] | Confidence: Very High
- Q17.** Which of the following is NOT a test item in the SAI Khelo India fitness battery for the 5-8 years age group? (A) BMI (B) Flamingo Balance Test (C) 50m Dash (D) Plate Tapping Test [Practice] | Confidence: Very High
- Q18.** The Rockport One-Mile Walk Test is primarily used to estimate an individual's: (A) $\dot{V}O_2$ Max (Cardiovascular fitness) (B) Flexibility (C) Muscular Endurance (D) Agility [PYQ] | Confidence: High
- Q19.** The Back Scratch Test for senior citizens evaluates: (A) Lower body flexibility (B) Upper body flexibility (C) Core strength (D) Balance [PYQ] | Confidence: Very High
- Q20.** What does BMI stand for? [PYQ] | Confidence: Very High

2-Mark

- Q21.** Mention the fitness index score formula (Long Method) and outline its norms for the Harvard Step Test. [PYQ] | Confidence: Very High

- Q22.** Write down the objectives and administration procedure of the Flamingo Balance Test. [PYQ 2024] | Confidence: Very High
- Q23.** What are the specific test items included in the SAI Khelo India fitness test battery for the age group 5-8 years (Class 1-3)? [Practice] | Confidence: Very High
- Q24.** Mention the purpose and the equipment required to conduct the Plate Tapping Test. [Practice] | Confidence: High
- Q25.** State the purpose and briefly explain the procedure of the '8-Foot Up and Go' test. [Practice] | Confidence: High
- Q26.** What is the Rockport One-Mile Walk Test? Mention the formula used to calculate $\dot{V}O_2$ max in this test. [PYQ] | Confidence: Medium
- Q27.** Explain the procedure of the Arm Curl Test for senior citizens. Mention the dumbbell weights for men and women. [PYQ] | Confidence: Very High
- Q28.** How is the Partial Curl-Up test administered? What does it measure? [Practice] | Confidence: High
- Q29.** Differentiate between the Chair Sit & Reach test and the Back Scratch test in terms of their objectives. [Practice] | Confidence: High
- Q30.** Write the purpose and equipment required for the 50m Dash (Standing Start) test. [PYQ 2024] | Confidence: Very High
- Q31.** Name any four items of the test battery of the AAPHER motor fitness test. [PYQ] | Confidence: Medium
- Q32.** State the purpose and the scoring procedure of the Chair Stand Test for senior citizens. [PYQ] | Confidence: High

3-Mark

- Q33.** Explain the procedure of any one strength test of the SAI Khelo India Fitness Test used for 9 to 18 years / class 4 to 12 children. [PYQ 2024] | Confidence: Very High
- Q34.** Detail the procedure, required equipment, and scoring of the Harvard Step Test. [PYQ] | Confidence: Very High
- Q35.** What is the SAI Khelo India fitness test? List all the fitness components and their respective tests for the 9-18 years age group. [PYQ] | Confidence: High
- Q36.** Explain the procedure of the Push-up test for boys and the Modified Push-up test for girls. What do they measure? [PYQ] | Confidence: Very High
- Q37.** A male student performed the Harvard Step Test for 4 minutes. His pulse was recorded during the recovery periods as 58 beats in the first half-minute, 50 beats in the second, and 42 beats in the third. Using the Harvard Step Test formula, calculate his Fitness Index. [PYQ 2025] | Confidence: High
- Q38.** Describe the objective, procedure, and scoring of the 'Sit and Reach' test. [Practice] | Confidence: High
- Q39.** How are lower body flexibility and upper body flexibility measured in senior citizens? Briefly explain the specific tests used for both. [Practice] | Confidence: High
- Q40.** Briefly explain the Johnsen-Methney test of motor educability and list its four test items. [PYQ] | Confidence: Medium
- Q41.** State the procedure and scoring system for the 6-Minute Walk test in the Rikli and Jones battery. [Practice] | Confidence: Very High
- Q42.** Explain the Barrow Motor Ability Test and list its three components. [PYQ 2018] | Confidence: Medium

5-Mark (Long Answer/Case-Study)

- Q43.** Write in detail the procedure and scoring of the 'Flamingo Balance Test' and 'Plate Tapping Test' recommended for the age group 5–8 years by SAI Khelo India. [PYQ 2024] | Confidence: Very High
- Q44.** What is the Rikli & Jones Senior Citizen Fitness Test? Detail the administration, procedure, and scoring of any three test items from this battery. [PYQ] | Confidence: Very High
- Q45.** Discuss the purpose of the Harvard Step Test. Explain its procedure, equipment required, and the formulas (Short and Long methods) for calculating the Fitness Index. [PYQ] | Confidence: Very High
- Q46. Case Study:** Mr. Lakshman, aged 65 years, worked as a civil engineer in a construction company. He had to walk and climb a lot as part of his job. After retirement, he settled with his son, spending time with his grandchildren. Recently, he is experiencing difficulty in doing certain chores which involve physical movement. His doctor advises a fitness evaluation. (i) Which specific test battery would you recommend to check Mr. Lakshman's fitness? (ii) How many total test items are there in the prescribed fitness test battery for Mr. Lakshman? (iii) The 'Chair sit & reach test' is done to check __. **(iv) The '8-foot up & go test' is performed to assess __.** [PYQ] | Confidence: Very High
- Q47. Case Study:** A primary school PE teacher is conducting a fitness assessment for students of Classes 1 to 3 (age group 5-8 years) under the SAI Khelo India Fitness Test guidelines to track their motor skill development early on. (i) Which specific test from this battery should the teacher use to measure the static balance of the young children? (ii) Which test is performed

to measure the coordination and speed of the upper limbs? (iii) To calculate a child's BMI, what two physical measurements must the teacher record? (iv) Mention the exact duration of the Plate Tapping Test in seconds. [Practice] | Confidence: High

Q48. Write in detail about the AAPER (American Alliance For Health, Physical Education and Recreation) Motor Fitness Test, listing all its components and their specific purposes. [PYQ] | Confidence: Medium

Q49. Discuss the administration of tests used to measure Cardiovascular Endurance and Speed in the SAI Khelo India battery for the 9-18 years age group. Write their detailed procedures. [PYQ 2024] | Confidence: High

Q50. Case Study: Rudra is working on a project to collect data for assessing Physical Fitness amongst Senior Citizens at his residential complex. He plans to administer tests for assessing Lower body Flexibility, Upper Body Flexibility, and Lower Body Strength. (i) List the specific test he should conduct for Lower Body Flexibility. (ii) List the specific test he should conduct for Upper Body Flexibility. (iii) List the specific test he should conduct for Lower Body Strength. (iv) Explain the procedure and scoring of the test used for Lower Body Strength. [PYQ] | Confidence: Very High

Answer Key

Q1. (A) Brouha **Q2.** (B) 5 pounds (8 lbs for men, 5 lbs for women) **Q3.** (A) $\frac{100 \times \text{Test duration in seconds}}{2 \times \text{Sum of heart beats in recovery periods}}$ **Q4.** (C) Flamingo Balance Test **Q5.** (A) Agility and dynamic balance **Q6.** (C) 6 inches **Q7.** (A) Lower body flexibility **Q8.** (A) 600m Run/Walk **Q9.** (A) To measure speed and agility **Q10.** (C) 15×2 feet **Q11.** (A) 45 cm (16 inches) (Note: for men it is 50cm / 20 inches) **Q12.** (C) Sit and reach test **Q13.** (A) 3 times (at 1-1.5 min, 2-2.5 min, and 3-3.5 min) **Q14.** (A) I-2, II-1, III-4, IV-3 **Q15.** (D) (A) is false, but (R) is true (BMI measures body composition, not cardiovascular endurance). **Q16.** (A) Lower body strength **Q17.** (C) 50m Dash (Used for 9-18 yrs, not 5-8 yrs) **Q18.** (A) $\dot{V}O_2$ Max (Cardiovascular fitness) **Q19.** (B) Upper body flexibility **Q20.** Body Mass Index. **Q21.** Formula: $\frac{100 \times \text{Test Duration (sec)}}{2 \times (\text{Sum of 3 pulse counts})}$. Norms: >90 Excellent, 80-89 Good, 65-79 High Average, 55-64 Low Average, <55 Poor. **Q22.** Objective: To measure static balance. Procedure: Athlete stands on the longer axis of a wooden beam on a single leg. The free leg is flexed at the knee and the foot is held close to the buttocks. Time starts when the instructor lets go. Number of falls/losses of balance in 60 seconds is recorded. **Q23.** 1. BMI (Body Composition). 2. Flamingo Balance Test (Static Balance). 3. Plate Tapping Test (Coordination). **Q24.** Purpose: To measure the speed and coordination of upper limbs. Equipment: Table (adjustable height), two yellow discs (20cm diameter), a rectangle (30x20cm), and a stopwatch. **Q25.** Purpose: To measure speed, agility, and balance while moving. Procedure: Participant sits on a chair, stands up on the 'go' command, walks 8 feet around a cone, and returns to a seated position. Scored in seconds. **Q26.** It measures cardiovascular fitness ($\dot{V}O_2$ Max). Formula: $\dot{V}O_2 \text{ Max} = 132.853 - (0.0769 \times \text{Weight}) - (0.3877 \times \text{Age}) + (6.315 \times \text{Gender}) - (3.2649 \times \text{Time}) - (0.1565 \times \text{Heart Rate})$. (Gender: Male=1, Female=0). **Q27.** Purpose: Upper body strength. Procedure: Participant sits on a chair, holds a dumbbell (8 lbs for men, 5 lbs for women) and performs maximum bicep curls in 30 seconds. **Q28.** Purpose: Abdominal/Core strength and endurance. Procedure: Lie on back, knees bent, arms straight by sides. Curl up to move fingers across a 6-inch parallel strip on the mat. Maximum curl-ups in 30 seconds are counted. **Q29.** Chair Sit & Reach measures lower body flexibility (hamstrings), testing if hands can reach the toes while sitting. Back Scratch measures upper body flexibility (shoulders), testing if hands can touch behind the back. **Q30.** Purpose: To determine acceleration and speed. Equipment: Measuring tape, stopwatch, cone markers, flat running surface. **Q31.** 1. Pull-ups (boys) / Flexed arm hang (girls). 2. Flexed leg sit-ups. 3. Shuttle run. 4. Standing broad jump. **Q32.** Purpose: To assess lower body strength for senior citizens. Scoring: The maximum number of full stands completed in 30 seconds. **Q33.** Push-Up Test (Boys): Measures upper body strength/endurance. Procedure: Start in plank position, hands shoulder-width apart. Lower body until chest is close to the ground, then push up. Count maximum correct continuous repetitions. Or Partial Curl-Up for core strength. **Q34.** Procedure: Step up and down on a bench (20 inches men / 16 inches women) at 30 steps/minute for 5 minutes. Wait 1 min. Count pulses at 1-1.5 min, 2-2.5 min, and 3-3.5 min. Use formula to calculate fitness index. **Q35.** The SAI Khelo India test (9-18 yrs) includes: 1. BMI (Body Composition). 2. 50m Dash (Speed). 3. 600m Run/Walk (Endurance). 4. Sit & Reach (Flexibility). 5. Strength tests: Partial Curl-Up (Core) and Push-Ups/Modified Push-ups (Upper body). **Q36.** Boys Push-ups: Body straight, toes on floor. Lower chest and push up. Measures upper body strength. Girls Modified Push-ups: Knees on the floor instead of toes. Lower chest and push up. Measures upper body strength and endurance. **Q37.** Test Duration = 4 mins = 240 seconds. Pulse sums = $58 + 50 + 42 = 150$. Fitness Index = $\frac{100 \times 240}{2 \times 150} = \frac{24000}{300} = 80$. Score falls in the 'Good' (or High Average) category. **Q38.** Objective: Lower back and hamstring flexibility. Procedure: Sit with legs straight, feet flat against the testing box. Reach forward along the measuring scale as far as possible and hold for 2 seconds. Score is the furthest distance reached in cm. **Q39.** Lower body: Chair Sit and Reach Test (Sit on edge of chair, keep one leg straight, reach toward toes; score is distance between fingers and toes). Upper body: Back Scratch Test (One hand over shoulder, other behind back; try to touch fingers; score is distance or overlap of middle fingers). **Q40.** Measures motor educability. Layout is a 15×2 feet mat. Items: 1. Front Roll, 2. Back Roll, 3. Jumping Half-Turn, 4. Jumping Full-Turn. **Q41.** Procedure: Mark a rectangular course (e.g., 50 yards). Instruct the participant to walk as fast as possible for exactly 6 minutes around the course. Scoring: Total distance covered in 6 minutes is recorded to the nearest yard/meter. **Q42.** Measures general motor ability. Components: 1. Standing Broad Jump (explosive leg power). 2. Zig-Zag Run (agility and speed). 3. Medicine Ball Put (upper body explosive strength). **Q43.** Flamingo Balance Test: Assesses static balance. Stand on a beam on one leg, hold the other foot near buttocks. Count falls in 60s. Plate Tapping Test: Assesses upper limb speed/coordination. Tap two yellow discs alternately over a centre rectangle with preferred hand, while other hand stays in middle. Time taken for 25 cycles (50 taps) is recorded. **Q44.** Designed to assess functional

fitness in older adults (60+). Items: 1. Chair Stand (lower body strength - stands in 30s). 2. Arm Curl (upper body strength - bicep curls in 30s). 3. Chair Sit & Reach (lower flexibility). 4. Back Scratch (upper flexibility). 5. 8-foot up & go (agility). 6. 6-min walk (aerobic endurance). **Q45.** Purpose: Measure cardiovascular endurance/fitness. Equipment: Bench (20"/16" high), stopwatch, metronome (120 beats/min for 30 steps/min). Short Formula: $\frac{100 \times \text{Duration (s)}}{5.5 \times \text{Pulse (1-1.5 min)}}$. Long Formula:

$\frac{100 \times \text{Duration (s)}}{2 \times (\text{Pulse 1} + \text{Pulse 2} + \text{Pulse 3})}$. **Q46.** (i) Rikli & Jones Senior Citizen Fitness Test. (ii) 6 items. (iii) Lower body flexibility. (iv) Agility and dynamic balance. **Q47.** (i) Flamingo Balance Test. (ii) Plate Tapping Test. (iii) Height (in meters) and Weight (in kg). (iv) Not fixed duration; the time taken to complete exactly 25 cycles (50 taps) is recorded. **Q48.** AAPHER Test includes: 1. Pull-ups (boys)/Flexed arm hang (girls) for upper body strength. 2. Flexed leg sit-ups for abdominal strength. 3. Shuttle run ($4 \times 10\text{m}$) for agility/speed. 4. Standing broad jump for explosive leg power. 5. 50-yard dash for speed. 6. 600-yard run/walk for cardiovascular endurance. **Q49.** 600m Run/Walk: Conducted on a track. Participants run or walk to cover 600m in the shortest time. Score is time taken in mins and secs. 50m Dash: From a standing start, run 50m at top speed. Time is recorded in seconds to the nearest tenth. **Q50.** (i) Chair Sit & Reach Test. (ii) Back Scratch Test. (iii) Chair Stand Test. (iv) Chair Stand Test Procedure: Sit in the middle of a chair, feet flat on the floor, arms crossed over the chest. On the "go" command, rise to a full stand and sit back down. Scoring: Record the total number of complete stands made in 30 seconds.

CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).